

Aerodynamic Parameter Study of NACA Airfoils using OpenFOAM

Project 02: NACA Airfoil Simulations with OpenFOAM

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Abstract

This report presents a parameter study for a flow around a series of NACA airfoils utilizing OpenFOAM. An automated Python pipeline was developed to generate structured meshes, run incompressible fluid simulations using the Spalart-Allmaras turbulence model, and post-process the resulting aerodynamic forces. A grid convergence study was first conducted to determine an appropriate number of surface discretization points, alongside a time convergence verification. Subsequently, a parameter study investigated the effect of varying maximum camber (NACA 0412 through 8412) on lift and drag coefficients at a zero angle of attack. All project code and reproducible scripts are hosted on GitHub.

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1 Introduction

NACA airfoils represent a set of standardized, parameterized profiles widely used in aerodynamic engineering and computational fluid dynamics (CFD) studies. For the 4-digit series (MPTT), the maximum camber (M), position of maximum camber (P), and thickness (TT) are governed by simple geometric equations.

When conducting CFD simulations, determining the optimal spatial and temporal resolution is crucial for balancing computational cost with physical accuracy. This project aims to deploy an automated workflow to first establish a numerically converged mesh and time scale. Following this, the pipeline evaluates how systematic changes in the maximum camber affect the steady-state lift (C_l) and drag (C_d) coefficients at a 0° angle of attack.

2 Methodology and Implementation

2.1 Simulation Setup

The simulations were performed in OpenFOAM 13 utilizing the `incompressibleFluid` solver. Turbulence was modeled using the one-equation Spalart-Allmaras (SA) Reynolds-Averaged Navier-Stokes (RANS) model, which is robust and well-suited for external aerodynamic flows. The freestream conditions were set to an inflow velocity of $U_\infty = 30$ m/s and a kinematic viscosity of $\nu = 1.5 \times 10^{-5}$ m²/s. Time-stepping was set conservatively at $\Delta t = 0.0005$ s to ensure stability during the initial transient phases of the solver.

2.2 Automation and Meshing Pipeline

The entire study was automated via Python scripts that run commands for OpenFOAM utilities (`blockMesh`, `checkMesh`, and `foamRun`). The full source code can be found in the project repository [1].

The domain mesh was generated using an adapted `curiosityFluidsAirfoilMesher.py` script, which creates a structured C-grid around the provided airfoil surface coordinates. The airfoil points were dynamically generated using the NACA 4-digit equations.

Symmetric Airfoil

A known topological degeneracy occurs when meshing perfectly symmetric profiles (e.g., NACA 0412, where $M = 0$). The meshing algorithm builds boundary-layer blocks by extruding surface normals; at the trailing edge of a symmetric airfoil, the upper and lower points share $y = 0$, causing boundary blocks to collapse and produce negative-volume cells. This was handled programmatically in the pipeline by applying a microscopic vertical offset ($\pm 0.0005 \cdot c \cdot t$) to the trailing edge points. This breaks the geometric degeneracy without influencing the macroscopic aerodynamic results.

3 Grid and Time Convergence Study

To determine a suitable spatial resolution, a grid convergence study was executed for a fixed airfoil geometry (NACA 2412). The airfoil was discretized with $N \in \{20, 50, 100, 200\}$ points along the surface.

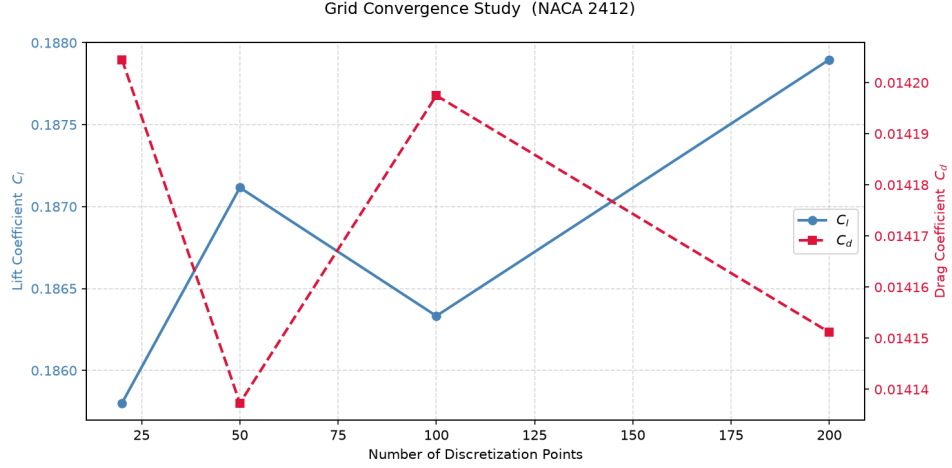
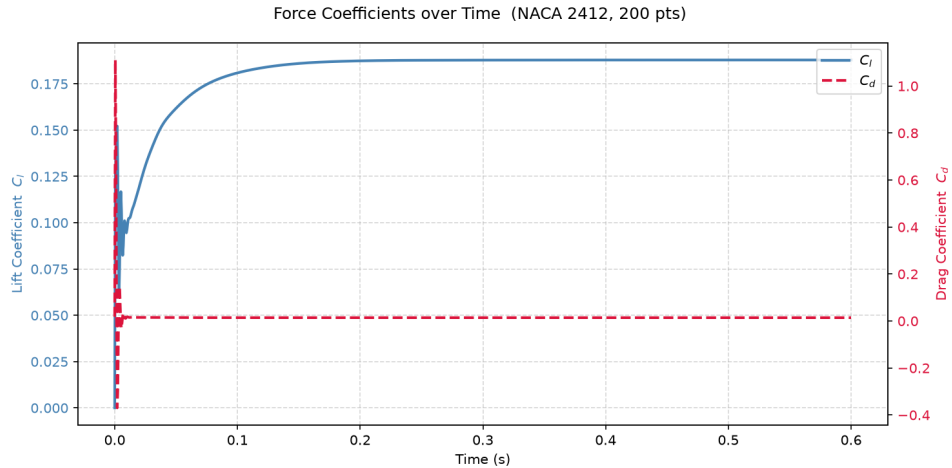


Figure 1: Grid convergence of Lift (C_l) and Drag (C_d) coefficients for the NACA 2412 airfoil as a function of the number of discretization points.

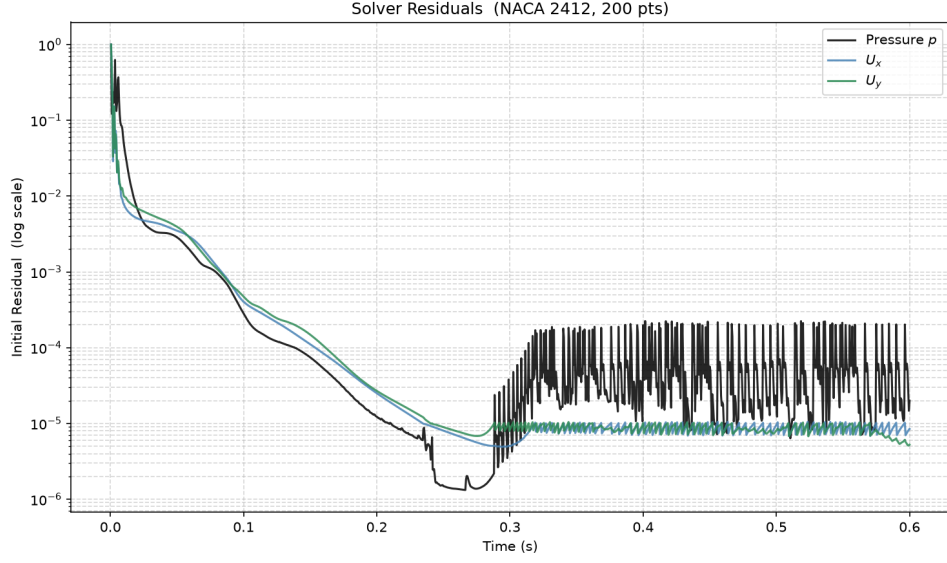
As shown in Figure 1, the force coefficients fluctuate under 1 % under the tested sets. A resolution of 100 points was deemed an acceptable compromise between numerical accuracy and computational efficiency for the subsequent parameter sweep.

To guarantee that the steady-state conditions were met, the force coefficients and solver residuals were tracked over time using the finest grid configuration ($N = 200$).



(a) Force coefficients over time.

Figure 2: Verification of steady-state convergence.



(b) Solver residuals over time.

Figure 2: Verification of steady-state convergence.

Figure 2 demonstrates that the initial transients decay rapidly. The lift and drag coefficients settle into constant values well before the final time step, and the pressure/velocity residuals drop sufficiently, confirming that $t = 0.6$ s is an adequate simulation time.

4 Parameter Study: Impact of Camber

Using the optimal configuration of 100 surface discretization points and an end time of 0.6 s, a parameter sweep was conducted to observe the isolated effect of airfoil camber. The maximum camber (represented by the ' M ' parameter) was varied from 0 to 8%. The position of maximum camber ($P = 4$) and maximum thickness ($TT = 12$) were held constant, effectively evaluating NACA 0412 through NACA 8412.

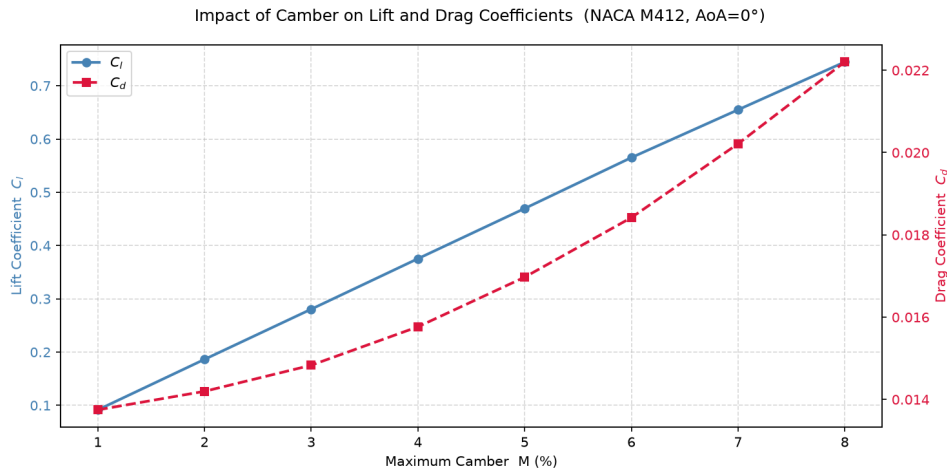


Figure 3: Impact of maximum camber (M) on steady-state Lift (C_l) and Drag (C_d) coefficients.

The results in Figure 3 shows the aerodynamic trend. As the camber of the airfoil increases, the lift coefficient (C_l) increases linearly. The drag coefficient (C_d) also increases, driven by the stronger adverse pressure gradients and the thicker wake profiles typical of highly cambered geometries at zero angle of attack.

5 Conclusion

This study successfully implemented an automated and reproducible OpenFOAM workflow in Python for 2D airfoil analysis. The convergence studies confirmed that sufficient surface discretization ($N \geq 100$) and simulation time are critical for stable and accurate aerodynamic load predictions. The automated parameter study correctly captured the expected physical trends, demonstrating that systematically increasing the maximum camber directly increases both the lift and drag coefficients of the airfoil. Future work could extend this automation pipeline to sweep across varying angles of attack (AoA) to automatically generate drag polar curves.

References

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- [2] RWTH Aachen University. *MSSD Teaching Repository*. <https://rwth-acom.pages.git.nrw/teaching/mssd/>
- [3] OpenFOAM Foundation. *OpenFOAM 13 User Guide*. <https://openfoam.org/>